SATHISHKUMAR R

rsathishkumar1963@gmail.com
linkedin.com/in/sathishkumar-r github.com/SathishKumar5115 Portfolio

Education

Vellore Institute of Technology Chennai **Expected 2027**
M.Tech in Computer Science and Engineering *CGPA: 9.58/10.0*

Shiv Nadar University Chennai **June 2025**
B.Tech in Computer Science and Engineering (IoT) *CGPA: 8.63/10.0*

Experience

Tata Consultancy Services (TCS) **May 2024 – July 2024**
IoT Engineer Intern *Chennai, India*

- Developed strategic IoT integration models for manufacturing environments, improving data collection efficiency by 30% through optimized sensor deployment and edge computing frameworks.
- Conducted comprehensive analysis of pharmaceutical and automobile manufacturing workflows, identifying 5+ process optimization opportunities that enhanced quality control accuracy by 25%.

AppLever Tech **Dec 2023 – Jan 2024**
Software Developer Intern *Hybrid, Bangalore, India*

- Engineered Flutter-based event management application with geofencing and Google Maps API integration, reducing event registration time by 40% and improving check-in accuracy by 50%.
- Implemented Firebase real-time database architecture supporting 1000+ concurrent users with 100% data synchronization reliability across Android and iOS platforms.

ISRO Propulsion Research Complex (IPRC) **June 2023 – July 2023**
In-Plant Trainee *Mahendragiri, India*

- Gained hands-on experience with enterprise-grade network infrastructure including VLAN segmentation, firewall configuration, and secure mail server deployment in high-security aerospace facilities.

Projects

Federated Learning Framework for NIDS | *Python, TensorFlow, Differential Privacy, Byzantine Aggregation* **Sep 2025 – Present**

- Developing privacy-preserving collaborative NIDS enabling organizations to train robust intrusion detection models without sharing sensitive network traffic, targeting 95% detection accuracy with 20% reduction in false positives through Byzantine-resilient aggregation and adaptive differential privacy mechanisms.

FoodIndex - Surplus Food Marketplace | *MERN Stack, MongoDB Geospatial, Node-Cron, React Context* **Jan 2026 – Feb 2026**

- Architected hyper-local food rescue platform with reverse Dutch auction pricing engine using MongoDB $geoNear for location-based discovery and automated cron jobs calculating dynamic pricing through 4-factor algorithm (40% time decay, 30% stock pressure, 20% demand, 10% urgency).

Sentinels Network Monitor | *Python, Flask, Scapy, SQLite, WebSockets* **Dec 2025**

- Engineered multi-threaded packet sniffer using Scapy to analyze live network traffic, implementing heuristic algorithms to detect SYN floods, UDP floods, and port scans with 92% accuracy.
- Designed interactive WebSocket-based dashboard to visualize protocol distribution and traffic volume, integrating IP geo-location APIs to map threat origins in real-time.

Mobile Applications Suite | *Flutter, Dart, Firebase, ESP32, MQTT, ML* **2023 – 2025**

- **SecureWare:** Designed IoT warehouse intrusion detection system using ESP32 microcontrollers and MQTT protocol for real-time intrusion detection, improving security response times by 40% via instant mobile alerts.
- **Air Sense:** Built scalable air quality monitoring system integrating ESP32 sensors with gamified Flutter interface, increasing user engagement by 62% through real-time insights.

Discord Bot Ecosystem | *Python, Discord.py, MongoDB, Flask* **June 2022 – Aug 2023**

- Created a multi-functional bot serving 6,000+ users across 20+ servers with virtual economy and moderation tools, handling 50,000+ daily commands with 99.8% uptime on Heroku.

Technical Skills

Programming Languages: Python, JavaScript, Java, C, Dart, SQL, HTML/CSS
Frameworks & Libraries: React.js, Node.js, Express.js, Flask, Flutter
Cloud & Databases: Firebase, MongoDB, MySQL, AWS
Developer Tools: Git, GitHub, Docker, VS Code,Postman, Android Studio, Figma